Group II, Row IV: extraction of the BFKL and finite terms from the left-over expression

1 The left-over expression and its check

After the JIMWLK term and the DGLAP + RC terms have been extracted, the left-over is

$$\begin{aligned}\frac{d^3N_{\rm left}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^4}\,\frac{1}{k^+}\,\log\Big(1-\frac{k^+}{\vee}\Big)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\,\Big[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\Big]\\ &\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^{e}(x)\\ &+\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]\\ &\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^{e}(x)\,. \qquad (1)\end{aligned}$$

Write $C\equiv\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}$ (both prefactors of (1) equal C, since $\frac{g^2}{4\pi^2}\cdot\frac{g^2}{2\pi^2}N_c=\frac{g^4N_c}{8\pi^4}$), $L_\epsilon=\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}$, and $\bar\xi=\frac{k^+}{\vee}$.

The left-over is correct, and its two terms are identified as follows.

(a) The first term is the *endpoint piece* $-C\,h(1)\,\log(1-\bar\xi)$ of the soft channel, $h(1)=\big[\frac{d_\perp}{2}L_\epsilon+1\big]e^{-ik(y'-y)}=L_\epsilon\,e^{-ik(y'-y)}$: the full endpoint logarithm was $\log\frac{(\vee-k^+)k^+}{\vee\Lambda}=\log\frac{k^+}{\Lambda}+\log(1-\bar\xi)$, the $\log\frac{k^+}{\Lambda}$ part left with the JIMWLK term, and $\log(1-\bar\xi)$ stayed. (Checked numerically: soft channel minus JIMWLK minus the locally subtracted integral equals $-C\,h(1)\log(1-\bar\xi)$ to $O(\Lambda)$.)

Convention remark. This term is present because the DGLAP term was extracted in the *locally subtracted* plus-prescription convention, $\int_{\bar\xi}^1 d\xi\,\frac{h(\xi)-h(1)}{1-\xi}$. If instead the *standard* plus distribution on $[0,1]$ is used in $\mathcal{P}_{gg}$ — the convention in which $\mathcal{P}_{gg,+}=2N_c\big[\xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{(1-\xi)_+}\big]$ is the textbook splitting function — then $\int_{\bar\xi}^1\frac{h}{(1-\xi)_+}=\int_{\bar\xi}^1\frac{h-h(1)}{1-\xi}+h(1)\log(1-\bar\xi)$ absorbs this first term into the DGLAP integral, and it disappears from the finite block. The two bookkeepings differ by exactly this one term; nothing else changes.

(b) The second term is the *azimuthal* term: the L_ϵ-free part of the regrouped expression, $+C\int_{\bar\xi}^1 d\xi\,\frac{1-\xi}{\xi}\big[\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')-d_\perp\frac{(\mathcal{K}\cdot r)(\mathcal{K}'\cdot r)}{r^2}\big]f(\xi)$, $r=y'-y$, with $d_\perp\to2$ (legitimate, since no pole multiplies it) and the upper limit already extended to 1 at a cost $O(\Lambda)$.

2 Step 1: no rapidity logarithm hides in the left-over

Before anything can be "extracted", note that *neither term of* (1) *contains a* $\log\frac{\vee}{k^+}$:

(i) The coefficient of the first term is $\log(1-\frac{k^+}{\vee})$, which tends to 0 as $\vee\to\infty$ at fixed k^+. It does not grow with the rapidity interval; it is a finite endpoint (convention) term, not a rapidity logarithm.

(ii) The second term is finite as $\bar\xi\to0$. Two mechanisms protect it. First, at fixed positions, substituting $u=\frac{1}{\xi}$ gives

$$\int_{\bar\xi}^1 d\xi\,\frac{1-\xi}{\xi}\,e^{-ik\cdot r/\xi}=\int_1^{1/\bar\xi}du\,\Big(\frac{1}{u}-\frac{1}{u^2}\Big)\,e^{-ik\cdot r\,u}\,, \qquad (2)$$

which *converges* as $\frac{1}{\bar\xi}=\frac{\vee}{k^+}\to\infty$ for $r\neq0$: the phase oscillates and the integral approaches a finite limit (checked numerically: the value stabilizes while $\log\frac{\vee}{k^+}$ would have grown without bound). Second, the dangerous collapse

region $r = O(\xi)$, where the phase stops oscillating, is doubly suppressed: writing $y' = y + \xi s$ costs a factor ξ^2 from the measure, and the azimuthal bracket has *exactly zero* angular average over the direction $\hat{r}$,

$$\int_0^{2\pi} \frac{d\theta_{\hat{r}}}{2\pi} \Big[\mathcal{K}\cdot\mathcal{K}' - 2\,(\mathcal{K}\cdot\hat{r})(\mathcal{K}'\cdot\hat{r}) \Big] = \mathcal{K}\cdot\mathcal{K}' - \mathcal{K}\cdot\mathcal{K}' = 0 \tag{3}$$

(in two dimensions $\langle 2\hat{r}^i\hat{r}^j\rangle_\theta = \delta^{ij}$).

Conclusion of Step 1. The BFKL logarithm cannot be found *inside* (1). It lives in the already-extracted DGLAP term: the soft channel $2N_c\frac{1-\xi}{\xi} \otimes L_\epsilon$ produces a hidden $\log \frac{\vee}{k^+}$ at its $\xi \to 0$ endpoint ($\xi \to 0$ is $p^+ \to \vee$: the unmeasured gluon carries almost all of the projectile momentum and the measured gluon is the soft one — the projectile-side rapidity region), and a rapidity logarithm multiplying L_ϵ is not physical. The BFKL term is therefore produced by an *exact add-and-subtract*: a subtraction term with the same $\xi \to 0$ limit is added to the finite block (where it protects the DGLAP soft channel), and its exactly computed value is subtracted back. Adding zero changes nothing, so the construction may equally well be carried out on the left-over (1), which is what we now do.

3 Step 2: constructing the subtraction term, explicitly

(2a) Two building blocks. Throughout, write

$$W(u,u') \equiv \Big[U^{ce'}(u') - U^{ce'}(x')\Big]\,[U^{ce}(u) - U^{ce}(x)]\,\rho^{e'}(x')\,\rho^{e}(x)\,, \tag{4}$$

so that the colour factor of (1) is $W(y,y')$, and define the ξ-dependent LO structure

$$\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,W(y,y') \tag{5}$$

($\xi = 1$ is the LO cross-section itself). The second building block is the dimensionally regularized transverse integral derived in the companion reduction note,

$$\int_z^{(d_\perp = 2-\epsilon)} K(y,y',z) = -\pi\,L_\epsilon\,, \qquad K(a,b,z) = \frac{(a-z)\cdot(b-z)}{(a-z)^2(b-z)^2}\,, \tag{6}$$

which we read *backwards*:

$$L_\epsilon = -\frac{1}{\pi}\int_z K(y,y',z)\,. \tag{7}$$

Eq. (7) trades the abstract ultraviolet bracket L_ϵ for a concrete transverse integral over one new point z. That is the whole trick of Step 2: once L_ϵ is a z-integral, a counterterm can be built under the *same* integral sign and compared with it point by point.

(2b) The term that needs a partner. The DGLAP term extracted before this note contains the soft channel

$$D_{\mathrm{soft}} \;=\; -\,\frac{g^2}{8\pi^2}\int_{\bar\xi}^1 \frac{d\xi}{\xi^2}\,2N_c\,\frac{1-\xi}{\xi}\,L_\epsilon\,\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}\,. \tag{8}$$

It has the two dangerous features at once: the ultraviolet bracket L_ϵ, and a hidden $\log\frac{\vee}{k^+}$ from its $\xi \to 0$ endpoint. A cross-section must not contain $\log\frac{\vee}{k^+} \times L_\epsilon$. The subtraction term is built to reproduce *exactly* the $\xi \to 0$ behaviour of (8) — nothing more — so that in the difference the rapidity logarithm and the μ-dependence disappear together.

(2c) Put D_{soft} under a z-integral. Insert (7) into (8) and write out (5):

$$D_{\mathrm{soft}} = +\,\frac{g^2}{8\pi^2}\int_{\bar\xi}^1 \frac{d\xi}{\xi^2}\,2N_c\,\frac{1-\xi}{\xi}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\frac{1}{\pi}\,K(y,y',z)\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,W(y,y')\,. \tag{9}$$

This is still exact; the overall sign flipped because of the minus sign in (7).

(2d) Fast and slow factors at $\xi \to 0$. The phase ties $y'-y$ to $O(\xi)$; concretely, in the variables $y = z + \xi(w-z)$, $y' = z + \xi(w'-z)$ (the exact map that Step 3 uses), $\xi \to 0$ at fixed (w,w',z) sends both parent points onto the point z: $y \to z$, $y' \to z$. The limit treats the factors of (9) very differently:

fast: $K(y,y',z) = \dfrac{K(w,w',z)}{\xi^2}$ blows up — this is what generates the logarithm;

slow: $\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y') \;\to\; \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z) = K(x,x',z)$ and $W(y,y') \to W(z,z)$ — smooth, finite limits.

(2e) Define the counterterm by freezing the slow factors. The subtraction term is obtained from (9) by *one* rule: keep the phase and the fast factor untouched, and replace the slow factors by their collapse-point values,

$$\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,W(y,y') \;\longrightarrow\; K(x,x',z)\,W(z,z)\,, \tag{10}$$

with an overall minus sign so that it subtracts. Explicitly, per ξ,

$$\frac{d^3N_\xi^{\rm sub}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\frac{1}{\pi}\,K(y,y',z)\,K(x,x',z)\,W(z,z)\,, \tag{11}$$

and, summed over ξ with the same measure, kernel weight and prefactor as (8),

$$S \;\equiv\; -\,\frac{g^2}{8\pi^2}\int_{\bar\xi}^1\frac{d\xi}{\xi^2}\,2N_c\,\frac{1-\xi}{\xi}\,\frac{d^3N_\xi^{\rm sub}}{d^3k} \;=\; -\,\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^5}\,\frac{1}{k^+}\int_{\bar\xi}^1\frac{d\xi}{\xi^2}\,\frac{1-\xi}{\xi}\int_{x,y,x',y',z} e^{-ik(y'-y)\frac{1}{\xi}}\,K(y,y',z)\,K(x,x',z)\,\Big[U^{ce'}(z)- \tag{12}$$

Nothing new has been invented: the $\frac{1}{\pi}$, the prefactors and the measure of (11)–(12) are all inherited from (9). Physically, (11) is the real-emission structure with the parent at the common point z — the two soft-emission (dipole) kernels $K(y,y',z)\,K(x,x',z)$ of amplitude and conjugate, and the Wilson lines at coincident points — precisely the building blocks of the real BFKL kernel. It is IR and UV finite in $d=2$ and contains no μ.

(2f) What the pair achieves. Adding (9) and (12), the two integrands sit under the same integral sign and combine into a single bracket:

$$D_{\rm soft}+S = +\,\frac{g^2}{8\pi^2}\int_{\bar\xi}^1\frac{d\xi}{\xi^2}\,2N_c\,\frac{1-\xi}{\xi}\,\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{K(y,y',z)}{\pi}\Big[\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,W(y,y')-K(x,x',z)\,W \tag{13}$$

By (2d) the square bracket vanishes *linearly* in ξ at fixed (w,w',z), and in those variables the z-integral converges at large z (the parent points travel with z, so both terms of the bracket fall off together); so the pair is free of $\frac{2}{\epsilon}$, γ_E and μ, and its soft region contributes $\int\frac{d\xi}{\xi}\times O(\xi)=$ finite — no $\log\frac{\vee}{k^+}$. The subtraction S alone therefore carries all the ultraviolet data and the entire rapidity logarithm of $D_{\rm soft}$. And S, unlike $D_{\rm soft}$, can be computed in closed form: that is Step 3.

4 Step 3: evaluating S in closed form, explicitly

(3a) Only the phase knows about ξ. Look at the inner integral of (12) at fixed ξ, with the ξ-independent factors pulled out of the y,y' integration:

$$\frac{d^3N_\xi^{\rm sub}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,x',z} K(x,x',z)\,W(z,z)\,\frac{1}{\pi}\int_{y,y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,K(y,y',z)\,. \tag{14}$$

The variable ξ appears in exactly one place: the phase. Strategy: change variables in the y,y' integral so that the phase loses its $\frac{1}{\xi}$; then every power of ξ is explicit, and the ξ-integral factorizes from the transverse one.

(3b) The substitution. At fixed z set

$$y = z+\xi\,(w-z) \iff w = z+\frac{y-z}{\xi}\,,\qquad y' = z+\xi\,(w'-z) \iff w' = z+\frac{y'-z}{\xi}\,, \tag{15}$$

an exact, invertible, linear change of variables — there is no approximation anywhere in Step 3.

(3c) Transform each factor, one at a time.

(i) Measure (two transverse dimensions each): $d^2y=\xi^2\,d^2w$, $d^2y'=\xi^2\,d^2w'$, hence $d^2y\,d^2y'=\xi^4\,d^2w\,d^2w'$.

(ii) Differences: $y-z=\xi\,(w-z)$, $y'-z=\xi\,(w'-z)$, and, subtracting the first from the second, $y'-y=\xi\,(w'-w)$.

(iii) Kernel:

$$K(y,y',z) = \frac{(y-z)\cdot(y'-z)}{(y-z)^2\,(y'-z)^2} = \frac{\xi^2\,(w-z)\cdot(w'-z)}{\xi^2(w-z)^2\;\xi^2(w'-z)^2} = \frac{1}{\xi^2}\,K(w,w',z)\,. \tag{16}$$

(iv) Phase:

$$e^{-ik\cdot(y'-y)\frac{1}{\xi}} = e^{-ik\cdot\xi(w'-w)\frac{1}{\xi}} = e^{-ik\cdot(w'-w)}\,. \tag{17}$$

The $\frac{1}{\xi}$ is eaten *exactly* — nothing is expanded or approximated.

(v) Spectators: $K(x,x',z)$, $W(z,z)$ and the x,x',z measure contain no y,y' and are untouched.

(3d) Assemble. Putting (i)–(v) into (14),

$$\frac{d^3N_\xi^{\rm sub}}{d^3k} = \xi^4\cdot\frac{1}{\xi^2}\cdot\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\frac{1}{\pi}\,\mathcal{I} = \xi^2\,\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\frac{1}{\pi}\,\mathcal{I}\,,\qquad \mathcal{I}\equiv\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,K(w,w',z)\,K(x,x',z)\,W(z,z)\,, \tag{18}$$

and $\mathcal{I}$ is manifestly ξ-independent: *all* the ξ-dependence of $\frac{d^3N_\xi^{\rm sub}}{d^3k}$ is the explicit factor ξ^2 (Jacobian ξ^4 times kernel rescaling ξ^{-2}).

(3e) The ξ-integral and the prefactor. Insert (18) into (12): the explicit $\frac{1}{\xi^2}$ of the measure is cancelled by the ξ^2 of (18), leaving

$$\int_{\bar\xi}^1 d\xi\,\frac{1-\xi}{\xi} = \int_{\bar\xi}^1\frac{d\xi}{\xi} - \int_{\bar\xi}^1 d\xi = \Big[\log\xi\Big]_{\bar\xi}^1 - \Big[\xi\Big]_{\bar\xi}^1 = \big(0-\log\bar\xi\big) - \big(1-\bar\xi\big) = \log\frac{\vee}{k^+} - \Big(1-\frac{k^+}{\vee}\Big) \tag{19}$$

(upper minus lower in both brackets; positive, as required for a positive integrand), and the prefactor assembles as

$$\frac{g^2}{8\pi^2}\cdot 2N_c\cdot\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\cdot\frac{1}{\pi} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\,. \tag{20}$$

Therefore

$$S = -\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\Big[\log\frac{\vee}{k^+} - \Big(1-\frac{k^+}{\vee}\Big)\Big]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,K(w,w',z)\,K(x,x',z)\,\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x) \tag{21}$$

5 Step 4: add and subtract, and the BFKL term

Insert $0 = S - S$ into (1), keeping S in its integral form (12) and replacing $-S$ by minus the closed form (21). Since (21) carries an overall minus sign, $-S$ is *plus* $\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\big[\log\frac{\vee}{k^+} - (1-\frac{k^+}{\vee})\big]\int KKW(z,z)$. Its $\log\frac{\vee}{k^+}$ part is the **BFKL contribution of this row**,

$$\boxed{\frac{d^3N^{\rm BFKL}_{\text{G-II, R IV}}}{d^3k} = +\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,K(w,w',z)\,K(x,x',z)\,\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)} \tag{22}$$

moved to the BFKL file (it appears there verbatim). Its coefficient is manifestly UV finite and μ-independent, as a BFKL kernel must be. The $-(1-\frac{k^+}{\vee})$ part of $-S$ is the boundary term of the finite block.

6 Result: the finite terms

Collecting everything,

$$\frac{d^3N_{\rm left}}{d^3k} = \frac{d^3N^{\rm BFKL}}{d^3k}\Big|_{(22)} + \frac{d^3N^{\rm finite}}{d^3k}\,, \tag{23}$$

with

$$\frac{d^3N^{\text{finite}}}{d^3k} = \underbrace{-\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\;L_\epsilon\;\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,/}_{\text{endpoint (plus-prescription convention) term}}$$

$$\underbrace{-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\frac{(1-\xi)}{\xi}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,K(y,y',z)\,K(x,x',z)\,\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x}_{\text{soft subtraction (= S)}}$$

$$\underbrace{-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\left(1-\frac{k^+}{\vee}\right)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,K(w,w',z)\,K(x,x',z)\,\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)]\,\rho^{e'}(x'}_{\text{boundary}}$$

$$\underbrace{+\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\right]}_{\text{azimuthal}} \tag{24}$$

Bookkeeping

(i) (soft subtraction S) + (boundary) + (BFKL entry (22)) = 0 identically: the construction is an exact add-and-subtract of (21), and all three objects are UV finite.

(ii) The role of S in the finite block is to pair with the soft channel $2N_c\frac{1-\xi}{\xi}\otimes L_\epsilon$ of the *already extracted* DGLAP term: per ξ, their sum is the μ-free combination (13), whose bracket is $O(\xi)$, so the pair contains no $\log\frac{\vee}{k^+}\times L_\epsilon$. The DGLAP term alone and S alone each contain that unphysical double structure; only their placement in different named blocks makes it look otherwise.

(iii) The endpoint term (first entry of (24)) is the difference between the two plus-prescription conventions. If $\mathcal{P}_{gg}$ is used with the standard $[0,1]$ plus distribution, this entry is absorbed into the DGLAP term and disappears from (24); nothing else changes.

(iv) (BFKL (22)) + (24) reproduces the left-over (1) identically — Steps 2–4 only added zero.